Model Architecture: gpt-oss-20b

Generated by sg_modelload

October 24, 2026

Contents

1 Model Overview **2**

2 Mathematical Representation **2**
- 2.1 The model as a composition of layers . . . 2
- 2.2 The layers, top to bottom . . . 3
- 2.3 Neurons and attention heads . . . 4
- 2.4 Architecture and parameters . . . 5
- 2.5 Output representations . . . 5
- 2.6 Intermediate states and sub-network decomposition . . . 5
- 2.7 Local linear representation . . . 6
- 2.8 Summary of notation . . . 6

3 Forward Pass in Pseudocode **6**

4 Compact Architecture (repeated layers written once) **7**
- 4.1 Modules outside the repeated stacks . . . 7
- 4.2 Language-model decoder: `model.layers` . . . 7

5 What Each Layer Does **8**

6 Parameter Count **9**

A Full Layer-by-Layer Table **9**

B Per-Layer Parameter Count **16**

1 Model Overview

Model name	gpt-oss-20b
Transformers class	`GptOssForCausalLM`
Path	`/models/gpt-oss-20b`
Total parameters	20,914,757,184 (20.91B)
Weight dtype	bfloat16
Checkpoint quantisation	mxfp4 (shapes below are the de-quantised ones)
Hidden size d	2,880
Decoder layers L	24
Query heads H	64
Key/value heads H_{kv}	8
Head dim d_h	64
MLP width d_{ff}	2,880
Experts E	32
Active experts per token	4
Vocabulary V	201,088
Max context length	131,072
Sliding window w	128
RMSNorm ϵ	1e-05
Tied input/output embeddings	False
Activation	SiLU
RoPE base θ	150,000

How to read this document. Section 2 treats the model mathematically, top-down: first the whole network as one composition $\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E$ of layers, then a definition of every layer down to single neurons and attention heads, then parameters, outputs, intermediate states and the local linear response. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Basic notation

B, T	batch size, sequence length (tokens)
d	hidden size (2880)
V	vocabulary size (201,088)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
RMSNorm(x)	$\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale γ)
LN(x)	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 Mathematical Representation

We describe the model *top-down*. First the whole network is written as one composition of functions (Section 2.1). Then every function that appears in that composition is defined (Section 2.2), down to individual neurons and attention heads. The remaining subsections describe the parameters, the outputs, the intermediate states, and the local linear behaviour.

2.1 The model as a composition of layers

Definition 2.1 (gpt-oss-20b as an L-layer decoder)**.** The model is a function $f_\theta : \mathcal{X} \to \mathcal{Y}$ with parameters θ. It maps a sequence of T token ids $x \in \mathcal{X} = \{1, \ldots, V\}^T$ to one logit vector per position, $y \in \mathcal{Y} = \mathbb{R}^{T \times V}$ ($V = 201{,}088$). It is a composition of an embedding E, $L = 24$ decoder blocks $\mathcal{B}^{(\ell)}$ and an output head Ψ:

$$f_\theta \;=\; \Psi \circ \mathcal{B}^{(L)} \circ \mathcal{B}^{(L-1)} \circ \cdots \circ \mathcal{B}^{(2)} \circ \mathcal{B}^{(1)} \circ E \tag{1}$$

Every block maps $\mathbb{R}^{T\times d} \to \mathbb{R}^{T\times d}$ ($d = 2880$), so the blocks can be stacked in any number. Block ℓ has its own parameters θ_ℓ and is stored in the checkpoint as `model.layers.`$(\ell - 1)$.

Definition 2.2 (Block types)**.** There are 2 block types (same function form within a type, different weights). A type map $c : \{1, \dots, L\} \to \{\mathrm{S}, \mathrm{F}\}$ says which type layer ℓ is: $\mathcal{B}^{(\ell)} = \mathcal{B}_{c(\ell)}(\,\cdot\,; \theta_\ell)$. Here $c = ((\mathrm{S}\,\mathrm{F})^{12})$, read left to right from layer 1. Written out, with the type as a subscript:

$$f_\theta = \Psi \circ \mathcal{B}_{\mathrm{F}}^{(24)} \circ \cdots \circ \mathcal{B}_{\mathrm{S}}^{(3)} \circ \mathcal{B}_{\mathrm{F}}^{(2)} \circ \mathcal{B}_{\mathrm{S}}^{(1)} \circ E \tag{2}$$

Grouping the repetitions with composition powers ($\mathcal{B}^{\circ n} = n$ consecutive blocks of the same type, each with its own weights) gives the compact form

$$f_\theta = \Psi \circ \Big(\big(\mathcal{B}_{\mathrm{F}} \circ \mathcal{B}_{\mathrm{S}} \big)^{\circ 12} \Big) \circ E \tag{3}$$

Layer pattern.

Symbol	Block type	Count	Params / block	Checkpoint indices
$\mathcal{B}_{\mathrm{S}}$	Sliding-window attention block	12	823,186,976	0, 2, …, 22 (every 2th)
$\mathcal{B}_{\mathrm{F}}$	Full (global) attention block	12	823,186,976	1, 3, …, 23 (every 2th)

Language-model decoder (24 layers). layers 0–23: $[\mathcal{B}_{\mathrm{S}}, \mathcal{B}_{\mathrm{F}}]$ repeated 12× (checkpoint indices, i.e. $\ell - 1$).

2.2 The layers, top to bottom

Every function in the composition above is defined here. Only the attention (or linear-attention) sub-layer mixes information *between* positions t. Every other sub-layer (norms, feed-forward networks, projections) acts on each token vector $h_t \in \mathbb{R}^d$ independently, e.g. $\mathrm{FFN}(h)_t = \mathrm{FFN}(h_t)$. All weights follow the convention $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, so a projection of a token vector is Wu (as in an affine layer $f_i(u) = A^{(i)}u + b^{(i)}$).

Definition 2.3 (Token embedding)**.** $E : \{1, \dots, V\}^T \to \mathbb{R}^{T\times d}$ looks up one row of the embedding matrix $W_E \in \mathbb{R}^{V\times d}$ per token:

$$E(x)_t = W_E[x_t, :], \qquad t = 1, \dots, T. \tag{4}$$

Definition 2.4 (Normalisation)**.** $\mathrm{RMSNorm}(u) = \gamma \odot u / \sqrt{\frac{1}{d}\sum_i u_i^2 + \epsilon}$, with a learned scale $\gamma \in \mathbb{R}^d$, applied to each token vector separately. It fixes the scale of a vector without changing its direction. Every norm in the model has its own γ, distinguished by a subscript (e.g. $\mathrm{RMSNorm}_1$, $\mathrm{RMSNorm}_{\text{final}}$).

Definition 2.5 (Block $\mathcal{B}_{\mathrm{S}}$: Sliding-window attention block)**.** A decoder block of this type maps $h \in \mathbb{R}^{T\times 2880}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id} + g)(h) = h + g(h)$:

$$\mathcal{B}_{\mathrm{S}} = (\mathrm{id} + \mathrm{FFN} \circ \mathrm{RMSNorm}_2) \circ (\mathrm{id} + \mathrm{Attn} \circ \mathrm{RMSNorm}_1) \tag{5}$$

Step by step, the sub-layers are:

$$\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
Q &= W_Q\, u + b_Q && H = 64 \text{ heads of size } d_h = 64\\
K,\ V &= W_K\, u + b_K, \quad W_V\, u + b_V && H_{kv} = 8 \text{ key/value heads}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\, Q_h K_{g(h)}^\top + M && \text{mask } M\text{: causal + sliding window of } w = 128\\
P_h &= \mathrm{softmax}\big([\, S_h \parallel z_h \,]\big)_{1:T} && \text{learned sink logit } z_h \text{ soaks up attention, then is dropped}\\
O_h &= P_h\, V_{g(h)} &&\\
\mathrm{Attn}(u) &= W_O\, [O_1; \dots; O_H] + b_O &&\\
\lambda &= W_r\, u + b_r \in \mathbb{R}^{32} && \text{router score per expert}\\
\mathcal{I} &= \mathrm{TopK}(\lambda,\ 4) && \text{pick 4 of 32 experts}\\
w &= \mathrm{softmax}(\lambda_{\mathcal{I}}) && \text{mixing weights}\\
[a,\, c] &= W_e^{gu}\, u + b_e^{gu} && \text{per expert } e\text{: gate } a, \text{ up } c\\
\mathrm{E}_e(u) &= W_e^{d}\big((c+1) \odot a\, \sigma(1.702\, a)\big) + b_e^{d} && \text{clamped SwiGLU expert}\\
\mathrm{FFN}(u) &= \textstyle\sum_{e\in\mathcal{I}} w_e\ \mathrm{E}_e(u) &&
\end{aligned}$$

Parameters of one block ($|\theta_\ell| = 823{,}186{,}976$): $W_Q \in \mathbb{R}^{4096\times 2880}$, $W_K \in \mathbb{R}^{512\times 2880}$, $W_V \in \mathbb{R}^{512\times 2880}$, $W_O \in \mathbb{R}^{2880\times 4096}$, $W_r \in \mathbb{R}^{32\times 2880}$, $W_e^{gu} \in \mathbb{R}^{5760\times 2880}$, $e = 1..32$, $W_e^{d} \in \mathbb{R}^{2880\times 2880}$, $e = 1..32$, plus 290,336 parameters in norm scales, biases and other vectors. This type is used 12 times.

Definition 2.6 (Block $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block)**.** A decoder block of this type maps $h \in \mathbb{R}^{T\times 2880}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id} + g)(h) = h + g(h)$:

$$\mathcal{B}_{\mathrm{F}} = (\mathrm{id} + \mathrm{FFN} \circ \mathrm{RMSNorm}_2) \circ (\mathrm{id} + \mathrm{Attn} \circ \mathrm{RMSNorm}_1) \tag{6}$$

Step by step, the sub-layers are:

$a = h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h))$	pre-norm + residual
$\mathcal{B}(h) = a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a))$	pre-norm + residual
$Q = W_Q\, u + b_Q$	$H = 64$ heads of size $d_h = 64$
$K,\ V = W_K\, u + b_K, \quad W_V\, u + b_V$	$H_{kv} = 8$ key/value heads
$Q,\ K = \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K)$	rotate by position
$S_h = s\, Q_h K_{g(h)}^{\top} + M$	mask M: causal
$P_h = \mathrm{softmax}\big([\, S_h \parallel z_h \,]\big)_{1:T}$	learned sink logit z_h soaks up attention, then is dropped
$O_h = P_h\, V_{g(h)}$	
$\mathrm{Attn}(u) = W_O\, [O_1; \ldots; O_H] + b_O$	
$\lambda = W_r\, u + b_r \in \mathbb{R}^{32}$	router score per expert
$\mathcal{I} = \mathrm{TopK}(\lambda,\ 4)$	pick 4 of 32 experts
$w = \mathrm{softmax}(\lambda_{\mathcal{I}})$	mixing weights
$[a,\ c] = W_e^{gu}\, u + b_e^{gu}$	per expert e: gate a, up c
$\mathrm{E}_e(u) = W_e^{d}\big((c+1) \odot a\, \sigma(1.702\, a)\big) + b_e^{d}$	clamped SwiGLU expert
$\mathrm{FFN}(u) = \sum_{e\in\mathcal{I}} w_e\ \mathrm{E}_e(u)$	

Parameters of one block ($|\theta_\ell| = 823{,}186{,}976$): $W_Q \in \mathbb{R}^{4096\times 2880}$, $W_K \in \mathbb{R}^{512\times 2880}$, $W_V \in \mathbb{R}^{512\times 2880}$, $W_O \in \mathbb{R}^{2880\times 4096}$, $W_r \in \mathbb{R}^{32\times 2880}$, $W_e^{gu} \in \mathbb{R}^{5760\times 2880}$, $e = 1..32$, $W_e^{d} \in \mathbb{R}^{2880\times 2880}$, $e = 1..32$, plus 290,336 parameters in norm scales, biases and other vectors. This type is used 12 times.

Definition 2.7 (Output head)**.** $\Psi : \mathbb{R}^{T\times d} \to \mathbb{R}^{T\times V}$ normalises the last hidden state and scores every vocabulary token with its own matrix $W_{\mathrm{LM}} \in \mathbb{R}^{V\times d}$:

$$\Psi(h)_t = W_{\mathrm{LM}}\ \mathrm{RMSNorm}_{\mathrm{final}}(h)_t \tag{7}$$

2.3 Neurons and attention heads

Definition 2.8 (Neuron, pre-activation value and state)**.** Each block has $E = 32$ experts, each with $d_{ff} = 2880$ neurons. Neuron j of expert e in layer ℓ is the scalar function of a normalised token vector $u \in \mathbb{R}^d$

$$\eta_{e,j}^{(\ell)}(u) = (c_j + 1)\, a_j\, \sigma(1.702\, a_j), \qquad a_j = W_{e,j}^{g,(\ell)} \cdot u + b_{e,j}^{g}, \quad c_j = W_{e,j}^{u,(\ell)} \cdot u + b_{e,j}^{u}, \tag{8}$$

where $W_{e,j}^{g}$, $W_{e,j}^{u}$ are the gate and up rows of W_e^{gu}. An expert's output is $\mathrm{E}_e(u) = \sum_j \eta_{e,j}(u)\, W_e^{d}[:, j] + b_e^{d}$. The network has $N = 2{,}211{,}840$ expert neurons, of which only 276,480 (the top-4 experts in every layer) are evaluated for a given token. For input x at position t:

- $\mathcal{V}(\eta; x, t) = a_j \in \mathbb{R}$ is the **pre-activation value** of neuron η (the argument of the non-linearity).
- $\mathcal{S}(\eta; x, t)$ is its **activation state**: *active* if $\mathcal{V} > 0$ and *inactive* if $\mathcal{V} \le 0$. The non-linearity is smooth, unlike ReLU, so an inactive neuron's output is strongly suppressed rather than exactly zero.

Definition 2.9 (Attention head and attention pattern)**.** Head h of an attention layer ℓ reads from the whole sequence through its attention pattern $P_h^{(\ell)}(x) \in \mathbb{R}^{T\times T}$ (row t = how much position t attends to each earlier position) and writes through its own $d \times d_h$ slice $W_{O,h}^{(\ell)}$ of W_O:

$$\mathrm{Attn}_\ell(u) = \sum_{h=1}^{H} P_h^{(\ell)}\, V_{g(h)}\, W_{O,h}^{(\ell)\top} + b_O. \tag{9}$$

$g(h)$ is the key/value head shared by query head h (grouped-query attention). The pattern P_h plays the role that the activation state plays for neurons: once P_h is fixed, the head is linear in its values. The network has 24 attention layers $\times$ $H = 64$ heads $= 1536$ attention heads.

2.4 Architecture and parameters

Definition 2.10 (Architecture and parameters)**.** The **architecture** is the tuple of sizes and the layer-type sequence:

$$\mathcal{A} = (V, d, L, c, H, H_{kv}, d_h, d_{ff}, E, k) = (201{,}088,\ 2880,\ 24,\ ((\mathrm{S\,F})^{12}),\ 64,\ 8,\ 64,\ 2880,\ 32,\ 4). \tag{10}$$

The **parameters** are $\theta = \{W_E,\ \theta_1, \ldots, \theta_L,\ \gamma_{\text{final}},\ W_{\text{LM}}\}$. Grouping the blocks by type,

$$\begin{aligned} |\theta| &= \sum_{\text{types } c} n_c\, |\theta_c| + |\theta_{\text{rest}}| \\ &= 12 \times 823{,}186{,}976 + 12 \times 823{,}186{,}976 + 1{,}158{,}269{,}760 \\ &= 20{,}914{,}757{,}184, \end{aligned}$$

where n_c is how often type c occurs and θ_{rest} holds everything outside the repeated stacks (embedding, final norm, ...; Table 3).

2.5 Output representations

Definition 2.11 (Output representations)**.** For an input sequence x, the output can be read at three levels:

1. **Logits:** $y = f_\theta(x) \in \mathbb{R}^{T \times V}$, one score per vocabulary token at every position.
2. **Next-token distribution:** $p(x)_{t,k} = \dfrac{\exp(y_{t,k})}{\sum_{j=1}^{V} \exp(y_{t,j})} = \Pr\big(x_{t+1} = k \mid x_1, \ldots, x_t\big)$.
3. **Predicted next token:** $z(x) = \operatorname{argmax}_{k \in \{1,\ldots,V\}} y_{T,k}$ (greedy decoding). Generation appends $z(x)$ (or a sample from $p(x)_T$) to x and applies f_θ again; in practice the keys and values of earlier positions are cached.

The model is **causal**: y_t depends only on $x_1, \ldots, x_t$, because every attention mask hides later positions (sliding-window layers additionally hide positions more than w tokens back). Training minimises the cross-entropy $-\sum_t \log p(x)_{t,x_{t+1}}$.

2.6 Intermediate states and sub-network decomposition

To study what happens at one layer $\ell \in \{1, \ldots, L\}$, split the composition into a prefix, the layer itself and a suffix:

$$f_\theta = \underbrace{\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(\ell+1)}}_{G_{\ell+1}\ \text{(suffix)}} \circ \underbrace{\mathcal{B}^{(\ell)}}_{\text{layer } \ell} \circ \underbrace{\mathcal{B}^{(\ell-1)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E}_{F_{\ell-1}\ \text{(prefix)}} \tag{11}$$

- **Prefix** $F_{\ell-1} : \mathcal{X} \to \mathbb{R}^{T \times d}$ maps the tokens to the hidden state entering layer ℓ ($F_0 = E$).
- **Hidden state** (residual stream) after layer ℓ: $h^{(\ell)} = F_\ell(x) = \mathcal{B}^{(\ell)}(h^{(\ell-1)}) \in \mathbb{R}^{T \times d}$, with $h^{(0)} = E(x)$.
- **Inside layer ℓ:** the attention write $a^{(\ell)} = \mathrm{Attn}_\ell\big(\mathrm{RMSNorm}_1(h^{(\ell-1)})\big)$, the mid-block state $\hat{h}^{(\ell)} = h^{(\ell-1)} + a^{(\ell)}$, the feed-forward input $u^{(\ell)} = \mathrm{RMSNorm}_2(\hat{h}^{(\ell)})$ and the feed-forward write $m^{(\ell)} = \mathrm{FFN}_\ell(u^{(\ell)})$.
- **Suffix** $G_{\ell+1} : \mathbb{R}^{T \times d} \to \mathbb{R}^{T \times V}$ maps $h^{(\ell)}$ to the logits ($G_{L+1} = \Psi$).

Residual stream. Because every sub-layer only *adds* to the hidden state, unrolling the composition gives the final hidden state as the embedding plus everything the layers wrote:

$$h^{(L)} = h^{(0)} + \sum_{\ell=1}^{L} \big(a^{(\ell)} + m^{(\ell)}\big) \tag{12}$$

So $f_\theta(x) = \Psi(h^{(L)})$: each layer reads the current stream and adds a correction to it.

2.7 Local linear representation

A ReLU network is piecewise affine: inside a polytope cell it collapses to one matrix. This model is not, because softmax, SiLU and the normalisations are smooth. Its local behaviour is described by the first-order (Jacobian) linearisation instead. Tokens are discrete, so we perturb the embedded input $h^{(0)}$.

Definition 2.12 (Local linear response)**.** Let $J_\ell = D\mathcal{B}^{(\ell)}(h^{(\ell-1)})$ be the Jacobian of block ℓ at the current hidden state (tensors flattened to vectors). The chain rule applied to the residual sub-layers $(\mathrm{id} + g_{\ell,i})$ gives

$$J_\ell = (I + Dg_{\ell,2})(I + Dg_{\ell,1}), \tag{13}$$

where $g_{\ell,1}$ is the attention branch, $g_{\ell,2}$ the feed-forward branch and each Dg is taken at that branch's input. For a small perturbation δ of $h^{(0)}$, the response of the network decomposed around layer ℓ is

$$f_\theta(h^{(0)} + \delta) - f_\theta(h^{(0)}) \approx G_x^{(\ell+1)} \cdot J_\ell \cdot F_x^{(\ell-1)} \cdot \delta, \tag{14}$$

with $F_x^{(\ell-1)} = J_{\ell-1} \cdots J_1$ (prefix) and $G_x^{(\ell+1)} = J_\Psi\, J_L \cdots J_{\ell+1}$ (suffix). Multiplying out the products of $(I + Dg)$ terms writes $J_\Psi J_L \cdots J_1$ as a sum over *paths*: the identity path straight from embedding to head, plus one term for every subset of sub-layers a signal passes through.

2.8 Summary of notation

Symbol	Meaning
$f_\theta : \mathcal{X} \to \mathcal{Y}$	the whole model, $\{1..V\}^T \to \mathbb{R}^{T\times V}$
$E,\ \Psi$	embedding and output head
$\mathcal{B}^{(\ell)},\ \theta_\ell$	block ℓ and its parameters, $\ell = 1, \ldots, L$
$\mathcal{B}_c,\ c(\ell)$	block type c; type of layer ℓ
$\mathcal{B}^{\circ n}$	n consecutive blocks of one type (each with its own weights)
$(\mathrm{id} + g)$	residual sub-layer $h \mapsto h + g(h)$
$h^{(\ell)}$	hidden state (residual stream) after layer ℓ; $h^{(0)}$ = embeddings
$a^{(\ell)},\ m^{(\ell)}$	what the attention / feed-forward sub-layer of layer ℓ adds to the stream
$\eta_j^{(\ell)}$	j-th feed-forward neuron of layer ℓ
$\mathcal{V}(\eta; x, t),\ \mathcal{S}(\eta; x, t)$	pre-activation value and active/inactive state of a neuron
$P_h^{(\ell)}$	attention pattern of head h in layer ℓ $(T \times T)$
$F_{\ell-1},\ G_{\ell+1}$	prefix (embedding to layer $\ell - 1$) and suffix (layer $\ell + 1$ to logits)
$J_\ell,\ F_x^{(\ell-1)},\ G_x^{(\ell+1)}$	Jacobian of block ℓ, and of the prefix / suffix at input x
$p(x),\ z(x)$	next-token distribution and greedy next-token prediction

3 Forward Pass in Pseudocode

Algorithm 1 gpt-oss-20b: full forward pass

Require: token ids x of shape $B \times T$

$h \leftarrow E(x)$ ▷ $h^{(0)}$: one vector per token, $B{\times}T{\times}2880$

$\ell \leftarrow 0$

for $r \leftarrow 1$ **to** 12 **do** ▷ layers 1–24: pattern of 2 blocks, repeated 12 times

  $\ell \leftarrow \ell + 1;\ h \leftarrow \mathcal{B}_{\mathrm{S}}(h;\, \theta_\ell)$ ▷ Sliding-window attention block

  $\ell \leftarrow \ell + 1;\ h \leftarrow \mathcal{B}_{\mathrm{F}}(h;\, \theta_\ell)$ ▷ Full (global) attention block

end for

$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$ ▷ $h = h^{(L)}$; this and the next step are Ψ

$y \leftarrow h\, W_{\mathrm{LM}}^{\top}$ ▷ logits $B{\times}T{\times}201,088$

return y ▷ softmax(y) = next-token probabilities

Algorithm 2 Block $\mathcal{B}_{\mathrm{S}}(h)$ — Sliding-window attention block

Require: h of shape $B \times T \times 2880$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top + b_Q$ ▷ $B{\times}T{\times}64{\times}64$ (64 query heads)
$K \leftarrow uW_K^\top + b_K;\ V \leftarrow uW_V^\top + b_V$ ▷ $B{\times}T{\times}8{\times}64$ (8 key/value heads, each shared by 8 query heads)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal + only the last 128 tokens; plus a learned sink column
$O \leftarrow PV$ ▷ $B{\times}T{\times}64{\times}64$
$m \leftarrow O\,W_O^\top + b_O$ ▷ concatenate heads, project back to $B{\times}T{\times}2880$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$\lambda \leftarrow uW_r^\top + b_r$ ▷ $B{\times}T{\times}32$ router scores
$\mathcal{I}, w \leftarrow \mathrm{TopK}(\lambda, 4),\ \mathrm{softmax}$ ▷ each token picks 4 of 32 experts
$m \leftarrow \sum_{e\in\mathcal{I}} w_e\ \mathrm{E}_e(u)$ ▷ expert: clamped SwiGLU, width 2880
$h \leftarrow r + m$ ▷ add residual
return h

Algorithm 3 Block $\mathcal{B}_{\mathrm{F}}(h)$ — Full (global) attention block

Require: h of shape $B \times T \times 2880$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top + b_Q$ ▷ $B{\times}T{\times}64{\times}64$ (64 query heads)
$K \leftarrow uW_K^\top + b_K;\ V \leftarrow uW_V^\top + b_V$ ▷ $B{\times}T{\times}8{\times}64$ (8 key/value heads, each shared by 8 query heads)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens; plus a learned sink column
$O \leftarrow PV$ ▷ $B{\times}T{\times}64{\times}64$
$m \leftarrow O\,W_O^\top + b_O$ ▷ concatenate heads, project back to $B{\times}T{\times}2880$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$\lambda \leftarrow uW_r^\top + b_r$ ▷ $B{\times}T{\times}32$ router scores
$\mathcal{I}, w \leftarrow \mathrm{TopK}(\lambda, 4),\ \mathrm{softmax}$ ▷ each token picks 4 of 32 experts
$m \leftarrow \sum_{e\in\mathcal{I}} w_e\ \mathrm{E}_e(u)$ ▷ expert: clamped SwiGLU, width 2880
$h \leftarrow r + m$ ▷ add residual
return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 3: Weights that appear once

Weight	Type	Shape	Params
`model.embed_tokens.weight`	Embedding	201088 × 2880	579,133,440
`model.norm.weight`	GptOssRMSNorm	2880	2,880
`lm_head.weight`	Linear	201088 × 2880	579,133,440

4.2 Language-model decoder: `model.layers`

Table 4: $\mathcal{B}_{\mathrm{S}}$: Sliding-window attention block — layers 0, 2, …, 22 (every 2th); weight names are relative to `model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.sinks`	GptOssAttention	64	64
`self_attn.q_proj.weight`	Linear	4096 × 2880	11,796,480
`self_attn.q_proj.bias`	Linear	4096	4,096
`self_attn.k_proj.weight`	Linear	512 × 2880	1,474,560
`self_attn.k_proj.bias`	Linear	512	512
`self_attn.v_proj.weight`	Linear	512 × 2880	1,474,560
`self_attn.v_proj.bias`	Linear	512	512
`self_attn.o_proj.weight`	Linear	2880 × 4096	11,796,480
`self_attn.o_proj.bias`	Linear	2880	2,880
`mlp.router.weight`	GptOssTopKRouter	32 × 2880	92,160
`mlp.router.bias`	GptOssTopKRouter	32	32
`mlp.experts.gate_up_proj`	GptOssExperts	32 × 2880 × 5760	530,841,600
`mlp.experts.gate_up_proj_bias`	GptOssExperts	32 × 5760	184,320
`mlp.experts.down_proj`	GptOssExperts	32 × 2880 × 2880	265,420,800
`mlp.experts.down_proj_bias`	GptOssExperts	32 × 2880	92,160
`input_layernorm.weight`	GptOssRMSNorm	2880	2,880
`post_attention_layernorm.weight`	GptOssRMSNorm	2880	2,880
Per block			**823,186,976**
× 12 blocks			**9,878,243,712**

Table 5: $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block — layers 1, 3, …, 23 (every 2th); weight names are relative to `model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.sinks`	GptOssAttention	64	64
`self_attn.q_proj.weight`	Linear	4096 × 2880	11,796,480
`self_attn.q_proj.bias`	Linear	4096	4,096
`self_attn.k_proj.weight`	Linear	512 × 2880	1,474,560
`self_attn.k_proj.bias`	Linear	512	512
`self_attn.v_proj.weight`	Linear	512 × 2880	1,474,560
`self_attn.v_proj.bias`	Linear	512	512
`self_attn.o_proj.weight`	Linear	2880 × 4096	11,796,480
`self_attn.o_proj.bias`	Linear	2880	2,880
`mlp.router.weight`	GptOssTopKRouter	32 × 2880	92,160
`mlp.router.bias`	GptOssTopKRouter	32	32
`mlp.experts.gate_up_proj`	GptOssExperts	32 × 2880 × 5760	530,841,600
`mlp.experts.gate_up_proj_bias`	GptOssExperts	32 × 5760	184,320
`mlp.experts.down_proj`	GptOssExperts	32 × 2880 × 2880	265,420,800
`mlp.experts.down_proj_bias`	GptOssExperts	32 × 2880	92,160
`input_layernorm.weight`	GptOssRMSNorm	2880	2,880
`post_attention_layernorm.weight`	GptOssRMSNorm	2880	2,880
Per block			**823,186,976**
× 12 blocks			**9,878,243,712**

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
LANGUAGE MODEL	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).

continued on next page

Layer (appears as)	What it does
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
LM head (output projection) `lm_head`	Maps each final hidden vector to one score (logit) per vocabulary token. Softmax of the logits gives the next-token probabilities. When tied, it reuses the embedding matrix.
Attention sinks `layers.self_attn`	One learned logit per head that joins the softmax as an extra "attend to nothing" slot. Lets a head put weight nowhere instead of being forced onto real tokens.
MoE router `mlp.router`	A small linear layer that scores every expert for every token. The top-k experts are run and their outputs mixed with softmax weights.
Expert gate+up projection (fused) `mlp.experts`	For each expert: one matrix producing both the gate and the up-projection (interleaved), plus a bias. Every expert is its own small MLP.
Expert down projection `mlp.experts`	For each expert: projects the gated hidden activations back to the model width (plus bias).
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP.
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.

6 Parameter Count

Table 7: Parameters by component

Component	Parameters	% of total
MLP / MoE	19,119,145,728	91.4%
Attention / token mixer	637,203,456	3.0%
Token embedding	579,133,440	2.8%
LM head	579,133,440	2.8%
Norms	141,120	0.0%
Total	**20,914,757,184**	**100.0%**

Table 8: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{S}}$ Sliding-window attention block	12	823,186,976	9,878,243,712
$\mathcal{B}_{\mathrm{F}}$ Full (global) attention block	12	823,186,976	9,878,243,712
Modules outside stacks	1	–	1,158,269,760
Total			**20,914,757,184**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
lm_head.weight	Linear	201088 × 2880	bfloat16	579,133,440
model	GptOssModel	–	–	–
model.embed_tokens.weight	Embedding	201088 × 2880	bfloat16	579,133,440
model.layers.0	GptOssDecoderLayer	–	–	–
model.layers.0.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.0.mlp	GptOssMLP	–	–	–
model.layers.0.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.0.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.0.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.0.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.0.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.0.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.0.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.0.self_attn	GptOssAttention	–	–	–
model.layers.0.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.0.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.0.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.0.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.0.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.0.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.0.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.0.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.0.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.1	GptOssDecoderLayer	–	–	–
model.layers.1.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.1.mlp	GptOssMLP	–	–	–
model.layers.1.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.1.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.1.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.1.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.1.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.1.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.1.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.1.self_attn	GptOssAttention	–	–	–
model.layers.1.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.1.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.1.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.1.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.1.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.1.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.1.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.1.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.1.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.2	GptOssDecoderLayer	–	–	–
model.layers.2.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.2.mlp	GptOssMLP	–	–	–
model.layers.2.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.2.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.2.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.2.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.2.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.2.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.2.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.2.self_attn	GptOssAttention	–	–	–
model.layers.2.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.2.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.2.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.2.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.2.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.2.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.2.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.2.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.2.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.3	GptOssDecoderLayer	–	–	–
model.layers.3.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.3.mlp	GptOssMLP	–	–	–
model.layers.3.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.3.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.3.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.3.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.3.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32

continued on next page

Layer path	Type	Shape	Dtype	Params
model.layers.3.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.3.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.3.self_attn	GptOssAttention	–	–	–
model.layers.3.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.3.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.3.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.3.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.3.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.3.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.3.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.3.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.3.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.4	GptOssDecoderLayer	–	–	–
model.layers.4.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.4.mlp	GptOssMLP	–	–	–
model.layers.4.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.4.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.4.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.4.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.4.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.4.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.4.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.4.self_attn	GptOssAttention	–	–	–
model.layers.4.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.4.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.4.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.4.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.4.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.4.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.4.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.4.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.4.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.5	GptOssDecoderLayer	–	–	–
model.layers.5.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.5.mlp	GptOssMLP	–	–	–
model.layers.5.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.5.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.5.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.5.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.5.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.5.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.5.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.5.self_attn	GptOssAttention	–	–	–
model.layers.5.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.5.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.5.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.5.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.5.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.5.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.5.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.5.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.5.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.6	GptOssDecoderLayer	–	–	–
model.layers.6.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.6.mlp	GptOssMLP	–	–	–
model.layers.6.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.6.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.6.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.6.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.6.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.6.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.6.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.6.self_attn	GptOssAttention	–	–	–
model.layers.6.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.6.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.6.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.6.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.6.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.6.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.6.self_attn.sinks	GptOssAttention	64	bfloat16	64

continued on next page

Layer path	Type	Shape	Dtype	Params
model.layers.6.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.6.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.7	GptOssDecoderLayer	–	–	–
model.layers.7.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.7.mlp	GptOssMLP	–	–	–
model.layers.7.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.7.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.7.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.7.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.7.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.7.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.7.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.7.self_attn	GptOssAttention	–	–	–
model.layers.7.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.7.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.7.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.7.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.7.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.7.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.7.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.7.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.7.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.8	GptOssDecoderLayer	–	–	–
model.layers.8.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.8.mlp	GptOssMLP	–	–	–
model.layers.8.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.8.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.8.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.8.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.8.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.8.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.8.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.8.self_attn	GptOssAttention	–	–	–
model.layers.8.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.8.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.8.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.8.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.8.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.8.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.8.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.8.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.8.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.9	GptOssDecoderLayer	–	–	–
model.layers.9.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.9.mlp	GptOssMLP	–	–	–
model.layers.9.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.9.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.9.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.9.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.9.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.9.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.9.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.9.self_attn	GptOssAttention	–	–	–
model.layers.9.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.9.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.9.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.9.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.9.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.9.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.9.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.9.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.9.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.10	GptOssDecoderLayer	–	–	–
model.layers.10.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.10.mlp	GptOssMLP	–	–	–
model.layers.10.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.10.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.10.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.10.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.10.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.10.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160

continued on next page

Layer path	Type	Shape	Dtype	Params
model.layers.10.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.10.self_attn	GptOssAttention	–	–	–
model.layers.10.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.10.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.10.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.10.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.10.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.10.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.10.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.10.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.10.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.11	GptOssDecoderLayer	–	–	–
model.layers.11.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.11.mlp	GptOssMLP	–	–	–
model.layers.11.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.11.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.11.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.11.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.11.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.11.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.11.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.11.self_attn	GptOssAttention	–	–	–
model.layers.11.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.11.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.11.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.11.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.11.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.11.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.11.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.11.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.11.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.12	GptOssDecoderLayer	–	–	–
model.layers.12.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.12.mlp	GptOssMLP	–	–	–
model.layers.12.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.12.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.12.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.12.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.12.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.12.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.12.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.12.self_attn	GptOssAttention	–	–	–
model.layers.12.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.12.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.12.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.12.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.12.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.12.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.12.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.12.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.12.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.13	GptOssDecoderLayer	–	–	–
model.layers.13.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.13.mlp	GptOssMLP	–	–	–
model.layers.13.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.13.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.13.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.13.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.13.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.13.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.13.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.13.self_attn	GptOssAttention	–	–	–
model.layers.13.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.13.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.13.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.13.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.13.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.13.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.13.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.13.self_attn.v_proj.bias	Linear	512	bfloat16	512

continued on next page

Layer path	Type	Shape	Dtype	Params
model.layers.13.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.14	GptOssDecoderLayer	–	–	–
model.layers.14.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.14.mlp	GptOssMLP	–	–	–
model.layers.14.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.14.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.14.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.14.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.14.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.14.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.14.post_attention_layernorm. weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.14.self_attn	GptOssAttention	–	–	–
model.layers.14.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.14.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.14.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.14.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.14.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.14.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.14.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.14.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.14.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.15	GptOssDecoderLayer	–	–	–
model.layers.15.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.15.mlp	GptOssMLP	–	–	–
model.layers.15.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.15.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.15.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.15.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.15.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.15.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.15.post_attention_layernorm. weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.15.self_attn	GptOssAttention	–	–	–
model.layers.15.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.15.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.15.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.15.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.15.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.15.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.15.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.15.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.15.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.16	GptOssDecoderLayer	–	–	–
model.layers.16.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.16.mlp	GptOssMLP	–	–	–
model.layers.16.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.16.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.16.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.16.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.16.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.16.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.16.post_attention_layernorm. weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.16.self_attn	GptOssAttention	–	–	–
model.layers.16.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.16.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.16.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.16.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.16.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.16.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.16.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.16.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.16.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.17	GptOssDecoderLayer	–	–	–
model.layers.17.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.17.mlp	GptOssMLP	–	–	–
model.layers.17.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.17.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.17.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.17.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.17.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.17.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160

continued on next page

Layer path	Type	Shape	Dtype	Params
model.layers.17.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.17.self_attn	GptOssAttention	–	–	–
model.layers.17.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.17.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.17.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.17.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.17.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.17.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.17.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.17.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.17.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.18	GptOssDecoderLayer	–	–	–
model.layers.18.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.18.mlp	GptOssMLP	–	–	–
model.layers.18.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.18.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.18.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.18.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.18.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.18.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.18.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.18.self_attn	GptOssAttention	–	–	–
model.layers.18.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.18.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.18.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.18.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.18.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.18.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.18.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.18.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.18.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.19	GptOssDecoderLayer	–	–	–
model.layers.19.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.19.mlp	GptOssMLP	–	–	–
model.layers.19.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.19.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.19.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.19.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.19.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.19.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.19.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.19.self_attn	GptOssAttention	–	–	–
model.layers.19.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.19.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.19.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.19.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.19.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.19.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.19.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.19.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.19.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.20	GptOssDecoderLayer	–	–	–
model.layers.20.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.20.mlp	GptOssMLP	–	–	–
model.layers.20.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.20.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.20.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.20.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.20.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.20.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.20.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.20.self_attn	GptOssAttention	–	–	–
model.layers.20.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.20.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.20.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.20.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.20.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.20.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.20.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.20.self_attn.v_proj.bias	Linear	512	bfloat16	512

continued on next page

Layer path	Type	Shape	Dtype	Params
model.layers.20.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.21	GptOssDecoderLayer	–	–	–
model.layers.21.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.21.mlp	GptOssMLP	–	–	–
model.layers.21.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.21.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.21.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.21.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.21.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.21.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.21.post_attention_layernorm. weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.21.self_attn	GptOssAttention	–	–	–
model.layers.21.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.21.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.21.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.21.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.21.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.21.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.21.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.21.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.21.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.22	GptOssDecoderLayer	–	–	–
model.layers.22.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.22.mlp	GptOssMLP	–	–	–
model.layers.22.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.22.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.22.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.22.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.22.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.22.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.22.post_attention_layernorm. weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.22.self_attn	GptOssAttention	–	–	–
model.layers.22.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.22.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.22.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.22.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.22.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.22.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.22.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.22.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.22.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.23	GptOssDecoderLayer	–	–	–
model.layers.23.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.23.mlp	GptOssMLP	–	–	–
model.layers.23.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.23.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.23.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.23.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.23.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.23.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.23.post_attention_layernorm. weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.23.self_attn	GptOssAttention	–	–	–
model.layers.23.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.23.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.23.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.23.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.23.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.23.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.23.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.23.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.23.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.norm.weight	GptOssRMSNorm	2880	bfloat16	2,880

Appendix B Per-Layer Parameter Count

Module	Params	Running total
lm_head	579,133,440	579,133,440

continued on next page

Module	Params	Running total
model.embed_tokens	579,133,440	1,158,266,880
model.layers.0.input_layernorm	2,880	1,158,269,760
model.layers.0.mlp.experts	796,538,880	1,954,808,640
model.layers.0.mlp.router	92,192	1,954,900,832
model.layers.0.post_attention_layernorm	2,880	1,954,903,712
model.layers.0.self_attn.k_proj	1,475,072	1,956,378,784
model.layers.0.self_attn.o_proj	11,799,360	1,968,178,144
model.layers.0.self_attn.q_proj	11,800,576	1,979,978,720
model.layers.0.self_attn	64	1,979,978,784
model.layers.0.self_attn.v_proj	1,475,072	1,981,453,856
model.layers.1.input_layernorm	2,880	1,981,456,736
model.layers.1.mlp.experts	796,538,880	2,777,995,616
model.layers.1.mlp.router	92,192	2,778,087,808
model.layers.1.post_attention_layernorm	2,880	2,778,090,688
model.layers.1.self_attn.k_proj	1,475,072	2,779,565,760
model.layers.1.self_attn.o_proj	11,799,360	2,791,365,120
model.layers.1.self_attn.q_proj	11,800,576	2,803,165,696
model.layers.1.self_attn	64	2,803,165,760
model.layers.1.self_attn.v_proj	1,475,072	2,804,640,832
model.layers.2.input_layernorm	2,880	2,804,643,712
model.layers.2.mlp.experts	796,538,880	3,601,182,592
model.layers.2.mlp.router	92,192	3,601,274,784
model.layers.2.post_attention_layernorm	2,880	3,601,277,664
model.layers.2.self_attn.k_proj	1,475,072	3,602,752,736
model.layers.2.self_attn.o_proj	11,799,360	3,614,552,096
model.layers.2.self_attn.q_proj	11,800,576	3,626,352,672
model.layers.2.self_attn	64	3,626,352,736
model.layers.2.self_attn.v_proj	1,475,072	3,627,827,808
model.layers.3.input_layernorm	2,880	3,627,830,688
model.layers.3.mlp.experts	796,538,880	4,424,369,568
model.layers.3.mlp.router	92,192	4,424,461,760
model.layers.3.post_attention_layernorm	2,880	4,424,464,640
model.layers.3.self_attn.k_proj	1,475,072	4,425,939,712
model.layers.3.self_attn.o_proj	11,799,360	4,437,739,072
model.layers.3.self_attn.q_proj	11,800,576	4,449,539,648
model.layers.3.self_attn	64	4,449,539,712
model.layers.3.self_attn.v_proj	1,475,072	4,451,014,784
model.layers.4.input_layernorm	2,880	4,451,017,664
model.layers.4.mlp.experts	796,538,880	5,247,556,544
model.layers.4.mlp.router	92,192	5,247,648,736
model.layers.4.post_attention_layernorm	2,880	5,247,651,616
model.layers.4.self_attn.k_proj	1,475,072	5,249,126,688
model.layers.4.self_attn.o_proj	11,799,360	5,260,926,048
model.layers.4.self_attn.q_proj	11,800,576	5,272,726,624
model.layers.4.self_attn	64	5,272,726,688
model.layers.4.self_attn.v_proj	1,475,072	5,274,201,760
model.layers.5.input_layernorm	2,880	5,274,204,640
model.layers.5.mlp.experts	796,538,880	6,070,743,520
model.layers.5.mlp.router	92,192	6,070,835,712
model.layers.5.post_attention_layernorm	2,880	6,070,838,592
model.layers.5.self_attn.k_proj	1,475,072	6,072,313,664
model.layers.5.self_attn.o_proj	11,799,360	6,084,113,024
model.layers.5.self_attn.q_proj	11,800,576	6,095,913,600
model.layers.5.self_attn	64	6,095,913,664
model.layers.5.self_attn.v_proj	1,475,072	6,097,388,736
model.layers.6.input_layernorm	2,880	6,097,391,616
model.layers.6.mlp.experts	796,538,880	6,893,930,496
model.layers.6.mlp.router	92,192	6,894,022,688

continued on next page

Module	Params	Running total
model.layers.6.post_attention_layernorm	2,880	6,894,025,568
model.layers.6.self_attn.k_proj	1,475,072	6,895,500,640
model.layers.6.self_attn.o_proj	11,799,360	6,907,300,000
model.layers.6.self_attn.q_proj	11,800,576	6,919,100,576
model.layers.6.self_attn	64	6,919,100,640
model.layers.6.self_attn.v_proj	1,475,072	6,920,575,712
model.layers.7.input_layernorm	2,880	6,920,578,592
model.layers.7.mlp.experts	796,538,880	7,717,117,472
model.layers.7.mlp.router	92,192	7,717,209,664
model.layers.7.post_attention_layernorm	2,880	7,717,212,544
model.layers.7.self_attn.k_proj	1,475,072	7,718,687,616
model.layers.7.self_attn.o_proj	11,799,360	7,730,486,976
model.layers.7.self_attn.q_proj	11,800,576	7,742,287,552
model.layers.7.self_attn	64	7,742,287,616
model.layers.7.self_attn.v_proj	1,475,072	7,743,762,688
model.layers.8.input_layernorm	2,880	7,743,765,568
model.layers.8.mlp.experts	796,538,880	8,540,304,448
model.layers.8.mlp.router	92,192	8,540,396,640
model.layers.8.post_attention_layernorm	2,880	8,540,399,520
model.layers.8.self_attn.k_proj	1,475,072	8,541,874,592
model.layers.8.self_attn.o_proj	11,799,360	8,553,673,952
model.layers.8.self_attn.q_proj	11,800,576	8,565,474,528
model.layers.8.self_attn	64	8,565,474,592
model.layers.8.self_attn.v_proj	1,475,072	8,566,949,664
model.layers.9.input_layernorm	2,880	8,566,952,544
model.layers.9.mlp.experts	796,538,880	9,363,491,424
model.layers.9.mlp.router	92,192	9,363,583,616
model.layers.9.post_attention_layernorm	2,880	9,363,586,496
model.layers.9.self_attn.k_proj	1,475,072	9,365,061,568
model.layers.9.self_attn.o_proj	11,799,360	9,376,860,928
model.layers.9.self_attn.q_proj	11,800,576	9,388,661,504
model.layers.9.self_attn	64	9,388,661,568
model.layers.9.self_attn.v_proj	1,475,072	9,390,136,640
model.layers.10.input_layernorm	2,880	9,390,139,520
model.layers.10.mlp.experts	796,538,880	10,186,678,400
model.layers.10.mlp.router	92,192	10,186,770,592
model.layers.10.post_attention_layernorm	2,880	10,186,773,472
model.layers.10.self_attn.k_proj	1,475,072	10,188,248,544
model.layers.10.self_attn.o_proj	11,799,360	10,200,047,904
model.layers.10.self_attn.q_proj	11,800,576	10,211,848,480
model.layers.10.self_attn	64	10,211,848,544
model.layers.10.self_attn.v_proj	1,475,072	10,213,323,616
model.layers.11.input_layernorm	2,880	10,213,326,496
model.layers.11.mlp.experts	796,538,880	11,009,865,376
model.layers.11.mlp.router	92,192	11,009,957,568
model.layers.11.post_attention_layernorm	2,880	11,009,960,448
model.layers.11.self_attn.k_proj	1,475,072	11,011,435,520
model.layers.11.self_attn.o_proj	11,799,360	11,023,234,880
model.layers.11.self_attn.q_proj	11,800,576	11,035,035,456
model.layers.11.self_attn	64	11,035,035,520
model.layers.11.self_attn.v_proj	1,475,072	11,036,510,592
model.layers.12.input_layernorm	2,880	11,036,513,472
model.layers.12.mlp.experts	796,538,880	11,833,052,352
model.layers.12.mlp.router	92,192	11,833,144,544
model.layers.12.post_attention_layernorm	2,880	11,833,147,424
model.layers.12.self_attn.k_proj	1,475,072	11,834,622,496
model.layers.12.self_attn.o_proj	11,799,360	11,846,421,856
model.layers.12.self_attn.q_proj	11,800,576	11,858,222,432

continued on next page

Module	Params	Running total
model.layers.12.self_attn	64	11,858,222,496
model.layers.12.self_attn.v_proj	1,475,072	11,859,697,568
model.layers.13.input_layernorm	2,880	11,859,700,448
model.layers.13.mlp.experts	796,538,880	12,656,239,328
model.layers.13.mlp.router	92,192	12,656,331,520
model.layers.13.post_attention_layernorm	2,880	12,656,334,400
model.layers.13.self_attn.k_proj	1,475,072	12,657,809,472
model.layers.13.self_attn.o_proj	11,799,360	12,669,608,832
model.layers.13.self_attn.q_proj	11,800,576	12,681,409,408
model.layers.13.self_attn	64	12,681,409,472
model.layers.13.self_attn.v_proj	1,475,072	12,682,884,544
model.layers.14.input_layernorm	2,880	12,682,887,424
model.layers.14.mlp.experts	796,538,880	13,479,426,304
model.layers.14.mlp.router	92,192	13,479,518,496
model.layers.14.post_attention_layernorm	2,880	13,479,521,376
model.layers.14.self_attn.k_proj	1,475,072	13,480,996,448
model.layers.14.self_attn.o_proj	11,799,360	13,492,795,808
model.layers.14.self_attn.q_proj	11,800,576	13,504,596,384
model.layers.14.self_attn	64	13,504,596,448
model.layers.14.self_attn.v_proj	1,475,072	13,506,071,520
model.layers.15.input_layernorm	2,880	13,506,074,400
model.layers.15.mlp.experts	796,538,880	14,302,613,280
model.layers.15.mlp.router	92,192	14,302,705,472
model.layers.15.post_attention_layernorm	2,880	14,302,708,352
model.layers.15.self_attn.k_proj	1,475,072	14,304,183,424
model.layers.15.self_attn.o_proj	11,799,360	14,315,982,784
model.layers.15.self_attn.q_proj	11,800,576	14,327,783,360
model.layers.15.self_attn	64	14,327,783,424
model.layers.15.self_attn.v_proj	1,475,072	14,329,258,496
model.layers.16.input_layernorm	2,880	14,329,261,376
model.layers.16.mlp.experts	796,538,880	15,125,800,256
model.layers.16.mlp.router	92,192	15,125,892,448
model.layers.16.post_attention_layernorm	2,880	15,125,895,328
model.layers.16.self_attn.k_proj	1,475,072	15,127,370,400
model.layers.16.self_attn.o_proj	11,799,360	15,139,169,760
model.layers.16.self_attn.q_proj	11,800,576	15,150,970,336
model.layers.16.self_attn	64	15,150,970,400
model.layers.16.self_attn.v_proj	1,475,072	15,152,445,472
model.layers.17.input_layernorm	2,880	15,152,448,352
model.layers.17.mlp.experts	796,538,880	15,948,987,232
model.layers.17.mlp.router	92,192	15,949,079,424
model.layers.17.post_attention_layernorm	2,880	15,949,082,304
model.layers.17.self_attn.k_proj	1,475,072	15,950,557,376
model.layers.17.self_attn.o_proj	11,799,360	15,962,356,736
model.layers.17.self_attn.q_proj	11,800,576	15,974,157,312
model.layers.17.self_attn	64	15,974,157,376
model.layers.17.self_attn.v_proj	1,475,072	15,975,632,448
model.layers.18.input_layernorm	2,880	15,975,635,328
model.layers.18.mlp.experts	796,538,880	16,772,174,208
model.layers.18.mlp.router	92,192	16,772,266,400
model.layers.18.post_attention_layernorm	2,880	16,772,269,280
model.layers.18.self_attn.k_proj	1,475,072	16,773,744,352
model.layers.18.self_attn.o_proj	11,799,360	16,785,543,712
model.layers.18.self_attn.q_proj	11,800,576	16,797,344,288
model.layers.18.self_attn	64	16,797,344,352
model.layers.18.self_attn.v_proj	1,475,072	16,798,819,424
model.layers.19.input_layernorm	2,880	16,798,822,304
model.layers.19.mlp.experts	796,538,880	17,595,361,184

continued on next page

Module	Params	Running total
model.layers.19.mlp.router	92,192	17,595,453,376
model.layers.19.post_attention_layernorm	2,880	17,595,456,256
model.layers.19.self_attn.k_proj	1,475,072	17,596,931,328
model.layers.19.self_attn.o_proj	11,799,360	17,608,730,688
model.layers.19.self_attn.q_proj	11,800,576	17,620,531,264
model.layers.19.self_attn	64	17,620,531,328
model.layers.19.self_attn.v_proj	1,475,072	17,622,006,400
model.layers.20.input_layernorm	2,880	17,622,009,280
model.layers.20.mlp.experts	796,538,880	18,418,548,160
model.layers.20.mlp.router	92,192	18,418,640,352
model.layers.20.post_attention_layernorm	2,880	18,418,643,232
model.layers.20.self_attn.k_proj	1,475,072	18,420,118,304
model.layers.20.self_attn.o_proj	11,799,360	18,431,917,664
model.layers.20.self_attn.q_proj	11,800,576	18,443,718,240
model.layers.20.self_attn	64	18,443,718,304
model.layers.20.self_attn.v_proj	1,475,072	18,445,193,376
model.layers.21.input_layernorm	2,880	18,445,196,256
model.layers.21.mlp.experts	796,538,880	19,241,735,136
model.layers.21.mlp.router	92,192	19,241,827,328
model.layers.21.post_attention_layernorm	2,880	19,241,830,208
model.layers.21.self_attn.k_proj	1,475,072	19,243,305,280
model.layers.21.self_attn.o_proj	11,799,360	19,255,104,640
model.layers.21.self_attn.q_proj	11,800,576	19,266,905,216
model.layers.21.self_attn	64	19,266,905,280
model.layers.21.self_attn.v_proj	1,475,072	19,268,380,352
model.layers.22.input_layernorm	2,880	19,268,383,232
model.layers.22.mlp.experts	796,538,880	20,064,922,112
model.layers.22.mlp.router	92,192	20,065,014,304
model.layers.22.post_attention_layernorm	2,880	20,065,017,184
model.layers.22.self_attn.k_proj	1,475,072	20,066,492,256
model.layers.22.self_attn.o_proj	11,799,360	20,078,291,616
model.layers.22.self_attn.q_proj	11,800,576	20,090,092,192
model.layers.22.self_attn	64	20,090,092,256
model.layers.22.self_attn.v_proj	1,475,072	20,091,567,328
model.layers.23.input_layernorm	2,880	20,091,570,208
model.layers.23.mlp.experts	796,538,880	20,888,109,088
model.layers.23.mlp.router	92,192	20,888,201,280
model.layers.23.post_attention_layernorm	2,880	20,888,204,160
model.layers.23.self_attn.k_proj	1,475,072	20,889,679,232
model.layers.23.self_attn.o_proj	11,799,360	20,901,478,592
model.layers.23.self_attn.q_proj	11,800,576	20,913,279,168
model.layers.23.self_attn	64	20,913,279,232
model.layers.23.self_attn.v_proj	1,475,072	20,914,754,304
model.norm	2,880	20,914,757,184